

Bitcoin Market Sentiment vs Trader Performance

Data Science Hiring Assignment

This study evaluates 211,224 Hyperliquid trade records against the daily Bitcoin Fear & Greed Index to identify regime-dependent trader behavior, performance patterns, and strategy implications.

Executive Summary

Fear produced the highest aggregate net PnL in the sample, while Extreme Greed delivered the highest win rate among trades with non-zero closed PnL. The strongest side asymmetry appeared during Extreme Greed, where SELL executions generated substantially more aggregate PnL than BUY executions. However, the sentiment score had only a weak positive Spearman association with daily-account PnL (approximately 0.066), indicating that sentiment is more useful as a market-regime filter than as a standalone predictive signal.

Data and Methodology

Trade timestamps were parsed from the IST timestamp field and normalized to calendar dates. Daily sentiment classification and score were joined by date. Net PnL was calculated as Closed PnL minus Fee. Win rate was calculated only on executions with non-zero Closed PnL, avoiding the incorrect classification of opening or position-building executions as losing trades. Aggregations were produced by sentiment, side, account, and daily-account observation.

Sentiment Regime Results

Regime	Trades	Net PnL	Win Rate*	Avg Size USD
Extreme Fear	21400	715,222	76.2%	5,350
Fear	61837	3,264,698	87.3%	7,816
Neutral	37686	1,253,546	82.4%	4,783
Greed	50303	2,087,031	76.9%	5,737
Extreme Greed	39992	2,688,141	89.2%	3,112

*Win rate is calculated only on non-zero Closed PnL trades.

Key Findings

- Fear led aggregate profitability.** Fear generated the highest aggregate net PnL, showing that adverse sentiment did not imply poor trader outcomes for this cohort.
- Extreme Greed had the strongest closing-trade hit rate.** The win rate on non-zero closed-PnL trades was highest in Extreme Greed, followed by Fear.
- Side asymmetry is economically meaningful.** During Extreme Greed, SELL executions generated approximately 2.53M gross PnL versus about 0.19M for BUY executions. This is the clearest regime-side interaction in the dataset.
- Sentiment alone is weak.** At the daily-account level, the Spearman correlation between sentiment score and PnL was only about 0.066. PnL was more strongly associated with activity and volume than with sentiment score.
- Account heterogeneity is substantial.** A small number of accounts contribute a large share of aggregate profit, while several accounts are net negative. Aggregate sentiment results should therefore be validated at account level.

Strategy Recommendations

Use sentiment as a regime filter rather than a standalone signal. In euphoric conditions, prioritize research on SELL/short setups, but require volatility-adjusted sizing and hard loss limits. In Fear regimes, maintain selective participation because aggregate opportunity is strong but account-level outcomes vary. Rank traders or strategy variants using rolling historical performance rather than full-sample results, and evaluate all strategies net of fees.

The next validation step should be a walk-forward, out-of-sample backtest using sentiment regime, side, normalized position size, volatility, and rolling account skill.

Limitations

This is observational analysis and does not establish causality. Closed PnL is recorded at execution level and many executions have zero Closed PnL, so win-rate definitions must be explicit. Results may be dominated by high-volume or highly profitable accounts. The analysis does not include external volatility, funding rates, order-book state, or Bitcoin return features. A production strategy requires temporal validation and risk-adjusted performance metrics.

Conclusion

The strongest practical insight is not that one sentiment label is universally best. Instead, trader performance depends on the interaction between market regime, trade side, activity, costs, and trader-specific skill. Sentiment is valuable context; the trading edge must come from conditional behavior and disciplined validation.